Matplotlib Lab- 14th March

Lab Instructions

1. Attempt **all the questions** using **Python and Matplotlib**.
2. Write the code in **Jupyter Notebook or Python script (.py file)**.
3. Each chart must include:
   - **Title**
   - **X-axis label**
   - **Y-axis label**
4. Use **appropriate datasets** (you may create sample data if not provided).
5. Use **different colors, markers, and styles** wherever applicable.
6. Use **proper variable names** in your program.

Lab Questions

1. Create a **Line Chart** to display the **monthly sales of a company for 12 months**.
2. Plot a **Multiple Line Chart** showing the **sales of three products (Product A, B, and C) for six months**.
3. Create a **Bar Chart** to show the **number of students enrolled in different courses** such as Python, Java, Data Science, and Web Development.
4. Generate a **Horizontal Bar Chart** to visualize the **population of five different cities**.
5. Create a **Stacked Bar Chart** showing the **number of male and female students in four departments**.
6. Draw a **Pie Chart** to represent the **market share of five smartphone brands**.
7. Create an **Exploded Pie Chart** showing **daily time spent on activities** like study, sleep, exercise, and entertainment.
8. Plot a **Histogram** to visualize the **distribution of marks obtained by 100 students**.
9. Create a **Scatter Plot** showing the relationship between **hours studied and marks obtained by students**.
10. Develop a **Bubble Chart** showing **population vs area of cities**, where bubble size represents **GDP**.
11. Plot an **Area Chart** showing the **yearly profit growth of a company over 6 years**.
12. Create a **Step Chart** to display **electricity consumption during different time intervals in a day**.
13. Generate a **Box Plot** to show the **salary distribution of employees in three different departments**.
14. Create a **Violin Plot** to visualize the **distribution of exam scores in different classes**.
15. Create **Subplots** in a single figure that display **four charts: Line Chart, Bar Chart, Pie Chart, and Scatter Plot**.